

PAPER

Bayesian Gene Set Benchmark Dose Estimation for “omic” responses

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Abstract

Motivation: Estimating a benchmark dose (BMD) as a reference point is a critical step in setting policy to regulate pollution and ensure safe environments. Toxicity can be measured for different endpoints, including changes in gene expression and histopathology for various tissues, and is typically explored one gene or tissue at a time in a univariate setting that ignores correlation. In this work we develop a multivariate estimation procedure to estimate the BMD for specified gene sets. Our approach extends the foundational univariate approach by accounting for correlation in a statistically principled way.

Results: We illustrate the method using data from a 5-day rat study and Hallmark gene sets and compare to existing BMD results computed by the EPA for both gene sets and apical histopathology endpoints. In contrast to previous ad-hoc methods, our principled approach provides the needed extension to bring the foundational univariate method into the multivariate world of transcriptomics. In addition to use in a regulatory setting, our method can provide hypothesis generation when gene sets correspond to mechanistic pathways.

Availability and Implementation: BS-BMD is implemented in R and C++ and will be available on github at <https://github.com/NIEHS/transcryptR>.

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Supplementary information: Supplementary data are available at *Journal Name*

Key words: Dose Response, Transcriptomic Response, Point of Departure

Introduction

Since the publication of the National Research Council report *Toxicity Testing in the 21st Century: A Vision and a Strategy* [Council et al., 2007], new alternative methodologies (NAMs) research, particularly “omic” technologies, has exploded. These methods typically focus around cellular assays that measure hundreds to thousands of features to better understand biological activity and quantify biological response to chemical exposure. For example, transcriptomic concentration-response analyses (e.g. [House et al., 2017, Ramaiahgari et al., 2019]) quantify the RNA transcript response by varying dose, elucidating molecular-level responses at different doses. These responses can then be evaluated to find a dose at which one is reasonably sure the measured response differs from no exposure, i.e., one is seeing a dose based change. This dose can be used as a point of departure (PoD) for low-dose extrapolation.

The standard method for estimating an apical endpoint PoD is based on a single concentration response curve, and is typically defined using the Benchmark Dose (BMD) (for

more information on this procedure for apical endpoints we refer the reader to Haber et al. [2018] and references therein). In our case, the measured biological effect is many, if not thousands of interacting responses. The BMDExpress approach [Yang et al., 2007, Phillips et al., 2019], which appeared early in the literature and is the current standard, analyzes each concentration-response independently by computing the PoD as the BMD defined for continuous responses [Crump, 1995], calculated using the US EPA’s BMDs software. To compute a PoD for a gene set, BMDExpress [Yang et al., 2007, Phillips et al., 2019] aggregates gene-level BMDs of the dose-responsive genes in each gene set (typically Gene Ontology Biological Process gene sets) and calculates a number of metrics as composite estimate of gene set potency. The metrics include mean, median and percentile-based BMD values. The most common approach to estimating and reporting a gene set potency and the one that is currently used in the EPA Transcriptional Assessment Product (ETAP) is the median BMD or BMDL value [Brennan et al., 2024].

The BMDExpress approach has several issues caused by model simplifications. Due to within-gene set gene correlations, it is possible to bias a gene's importance in the ontology set. Further, in choosing a single representative gene from a gene set, uncertainty from the contributions of the other genes is not incorporated into uncertainty estimates for a given set. This issue is also present in other methods which rely on individual BMD estimates that are then grouped and processed to arrive at a set-level BMD Dean et al. [2017], Thomas et al. [2007].

Nyffeler et al. [2021] attempted to correct BMDExpress' deficiencies by creating reduced rank feature sets for a given gene ontology. The data are reduced using principal component analysis (PCA) with an eigen-feature reduction at either the global or category level (i.e., based upon all available assay data or a transcriptomic data from a specific gene-ontology). The eigen-feature's data is used to compute the covariance matrix and the Mahalanobis distance which are used for dose-response modeling using tcplfit2 [Sheffield et al., 2022]. A similar approach by Basili et al. [2022] utilizes the Pathway-Level Information Extractor (PLIER) [Mao et al., 2019] to reduce data dimensionality by grouping genes into latent variables (LVs) based on their co-expression patterns. These LV scores were then modeled using tcplfit2 to estimate PoDs reflecting pathway-level activity. While these methods address the issue of gene correlations and aggregating dose responses, the use of PCA or eigendecomposition forces the user to only consider gene sets represented in an eigenvector or latent factor. Basili et al. [2022] include a prior to encourage specific gene sets in the latent factors, but the corresponding covariance then depends on the set of genes being considered, rather than being purely data dependent.

Considering these issues, we propose a new method called Bayesian Set Benchmark Dose analysis (BS-BMD). This approach estimates the BS-BMD for any set of genes, which can be defined using gene ontologies or by the analyst. Our strategy models the genes' dose-response curves simultaneously while estimating the covariance matrix between genes; we then construct an ontology BS-BMD. The computation reduces to Crump [1995] for univariate response but scales to sets that represent hundreds or even thousands of genes. Using the data of Gwinn et al. [2020], we show that the approach is more effective at determining biological activity than the current approaches and provides statistical bounds on the BMD estimate for each gene set.

System and Methods

Univariate BMD

For a single continuous response, the BMD is the gold standard used to define the PoD. Many use the BMD defined by Crump [1995]; here, the BMD is "the dose corresponding to a prescribed change in the mean response relative to the standard deviation, σ ." Crump specifies the "prescribed change" by changes in the probability of observing a response greater than or equal to a cut point at no exposure. It is this approach that we extend into multiple dimensions.

In Crump's approach, $\mu(d)$ is the dose-response function representing the mean response given dose d . He defines risk, i.e., the increased probability of an extreme response above background exposure as

$$Pr(Y > Y_0 | d = BMD) - Pr(Y > Y_0 | d = 0) = BMR, \quad (1)$$

where Y_0 is a cut point typically defined by a probability of occurrence at no exposure, e.g., 1%, and BMR is the change in the frequency of observing an event at least as extreme as Y_0 given the dose, BMD, e.g., 10%.

For Gaussian errors with constant variance, this corresponds to standardized changes in the mean, i.e.,

$$\frac{|\mu(d) - \mu(0)|}{\sigma} = \text{BMRF}, \quad (2)$$

where the quantity BMRF represents the number of standard deviations change of the response from no exposure. For example, assume a Gaussian response distribution and an "adverse response" defined as a response that would occur with a frequency of 1% or less under background exposure. If BMRF = 1.1, then a dose inducing a 1.1 standard deviation change in the mean response would correspond to the BMR being 10%. That is, if $\sigma = 1$ and $\mu(0) = 0$, then $Pr(Y > 2.33 | d = 0) = 0.01$, and the BMD is the dose that increases the mean by 1.1 standard deviations from background such that $Pr(Y > 2.33 | d = \text{BMD}) = 0.11$. This concept is illustrated in Figure 1.

In most applications, when the standard deviation definition of the BMD is used the BMRF is rounded down to be 1. For our purposes, we retain 1.1 because it matches with the original probability calculus of Crump of 1% and 10% for adverse and background response changes. Other values can be used instead of 1 and 10%.

Crump's BMD is defined using response probabilities. In practice, toxicologists do not interpret it in this manner. Instead, most understand the BMD as the dose inducing a specified change from the background, specifically by a standard-deviation change in the mean. This interpretation is essentially the BMD definition of Slob et al. [2005]. Confusion is understandable due to their numerical similarity; however, the latter approach is not generalizable for groups of genes. Thus, we retain the motivation from the original nomenclature to motivate our method.

BS-BMD

We extend the procedure described above to sets of dose response curves by creating a multivariate analog to (2). Our formulation aggregates individual mean dose-response differences while accounting for correlations between genes and can be distilled into a single test statistic.

To expand (2), let $\mu(d)$ be a vector function representing the dose-response for each gene response, and Σ be the covariance between genes. Our generalization of the BMD for a set of genes is the first dose satisfying

$$[\mu(\text{BMD}) - \mu(0)]^T \Sigma^{-1} [\mu(\text{BMD}) - \mu(0)] = T, \quad (3)$$

where T is a pivotal quantity computed based upon the probability of an adverse response, and can be related to the probability of an "adverse response" as in (2). This definition gives an "effect size" for an entire group analogous to the standard deviation change proposed by Crump [1995]. Though this test is a quadratic when compared to the original linear test, it reduces to Crump's approach for a set of size one where $\text{BMRF} = \sqrt{T}$. Using (3), we find the lowest dose solving the equality. If there is no such dose in the range of the tested dose, the BS-BMD is set to the maximum dose tested. We do this to indicate that that gene-set is non-responsive at the specified BMRF across the exposed dose range. The calculation of the threshold T follows the same principle to Crump [1995].

In what follows, we find the mean change in $\mu(d)$ corresponding to the probability of an extreme event increasing by 10%. The BS-BMD is computed for the entire set of interest and not a quantity specified for a single gene. This approach allows sets to be defined based upon biological understanding of gene sets (i.e., an ontology) while being less sensitive to experimental variability relating to a single gene response. It is not assumed or necessary that every gene in the set contribute to the aggregated statistic. Although unlikely, a gene set with only one active gene could feasibly have a low BMD due to the extreme response of that one gene.

Calculation of the BMR Threshold

Genes can be up- or down-regulated; consequently, there is no reason to expect a transcriptomic response to increase or decrease with exposure *a priori*. Our statistic considers extreme events in both tails of the distribution. The positive-valued quadratic test statistic T is based upon both negative and positive changes in the response. A BMR related to 10% using Crump [1995] is equivalent to a 20% change in response using our approach. Figure 2 illustrates the threshold for the 98th and 78th quantiles, derived from adding both tails of the normal distribution, i.e., $1 + 10 + 10 + 1 = 22\%$. In what follows, we investigate cutoffs to be consistent with current practice but note that different cutoffs may be desirable.

If the point-wise errors are Gaussian, then after subtracting the assumed mean, $\mu(0)$, and scaling by the inverse covariance, Σ^{-1} , as in Equation (3), one can apply standard normal theory and compute T from a χ_k^2 distribution, where k represents the degrees of freedom and is the rank of Σ . Using 2% as the background probability, and 20% as the increased frequency of response, one can compute T using the χ_k^2 distribution, i.e., $T = F^{-1}(0.98, k) - F^{-1}(0.78, k)$, where $F^{-1}(\cdot, k)$ is the inverse of the cumulative distribution function of a χ_k^2 distribution. This definition is equivalent to a univariate BMR for a standard Gaussian distribution, moving the mean 1.1 standard deviations in either direction. We use this assumption and thus estimate our pivotal quantity using the χ_k^2 distribution.

Bayesian Model

To compute the BS-BMD, we estimate $\mu(d)$ and Σ and do this from a Bayesian perspective using the following model for a set of genes:

$$Y_{ji} = \mu_i(d) + \lambda_i \eta_j + \epsilon_i,$$

where $\epsilon_i \sim N(0, \sigma_i^2)$, and the dose-response $\mu_j(d) = X\beta_i$ is computed using a spline basis defined from knots D_j , $1 \leq j \leq k$. The response Y_{ji} represents the response for observation j and gene i , $1 \leq i \leq g$. In this model, β_i is the vector of spline coefficients for gene i , λ_i is a $m \times 1$ row vector of latent factors for gene i , and η_j is the random effect for observation j . The correlation is modeled through the latent factor matrix using Λ , a $g \times m$ matrix whose rows are λ_i . We give the full Bayesian specification of these parameters in the appendix, but emphasize that $\eta \sim N(0, I_{m \times d})$, and marginalizing out η implies the vector Y has the following distribution:

$$Y | \Lambda \sim N(X\beta, \Lambda\Lambda^\top + \text{diag}[\{\sigma_1^2, \dots, \sigma_g^2\}]),$$

where $\text{diag}[\dots]$ is the diagonal matrix with diagonal elements $\{\sigma_1^2, \dots, \sigma_g^2\}$. Additionally, to model the covariance matrix, we pick m large and learn the number of factors automatically using a sparse latent factor model [Bhattacharya and Dunson, 2011]. Given an initial m possible factors, the sparsity inducing method shrinks unneeded factors to 0. Hence the matrix η has

dimension $m \times d$ and λ_i has dimension $1 \times m$, while the stacked version Λ will have dimension $g \times m$. Any remaining pointwise noise is captured in ϵ_i , which is assumed to have distribution $N(0, \tau^{-1}I_d)$.

Under this structure, one forms the test in Equation (3) letting $\Sigma = \Lambda\Lambda^\top + \text{diag}[\{\sigma_1^2, \dots, \sigma_g^2\}]$. To estimate all parameters, a Bayesian Markov Chain Monte Carlo (MCMC) procedure is developed and the details are left to the appendix. The BMD is computed in each iteration following the description in Section 2; this includes recomputing the null statistic conditional on the current parameter estimates.

The Null Distribution of the Dose-Response

Even when there is no dose-response, $\mu(d)$ will be estimated with variability rather than as a constant over d due to the nature of MCMC sampling. The test threshold T should therefore be modified to account for this additional variance and control false positives when computing the BS-BMD. From the model, one can remove the fitted mean and re-estimate $\mu(d)$ assuming no dose-response, which we define as $\mu^*(d)$. Under the Bayesian framework, one can estimate $\mu^*(d) | \mu(0), \epsilon$, where $\epsilon = \{\epsilon_1, \dots, \epsilon_g\}$, and then compute $T^* | \mu^*(0), \epsilon$, i.e.,

$$[\mu^*(d) - \mu^*(0)]^\top \Sigma^{-1} [\mu^*(d) - \mu^*(0)] = T^*(d). \quad (4)$$

This quantity is added to the critical value to form T . In practice the null response T^* is subtracted from Equation (3) to get a corrected response for the test statistic T :

$$[\mu(d) - \mu(0)]^\top \Sigma^{-1} [\mu(d) - \mu(0)] - T^*(d) = T \quad (5)$$

For full details of the computations, we refer the reader to the online supplement.

Data and Analysis

Gwinn et al. [2020] compares BMD estimates for sets of genes between 5 day and two year studies where rats are independently exposed by oral gavage to 18 chemicals. RNA transcripts are sampled from the liver and kidney using the TempO-Seq S1500+ platform, which measures approximately 2,700 probes. Each probe is designed to measure a specific gene and while different probes may correspond to the same gene grouping they can reasonably be assumed to equate a probe to a gene. Gwinn et al. [2020] define gene sets based on the Gene Ontology (GO) Biological Process [Ashburner et al., 2000, Aleksander et al., 2023], a dynamic, unified vocabulary and definition for eukaryotic gene and protein roles over a hierarchy of >10k biological processes. For simplicity we use the Hallmark gene sets [Liberzon et al., 2015] to illustrate the proposed method, but GO sets or any curated or computationally derived gene set could be used.

In Gwinn et al. [2020], chemicals vary in toxicity as measured by the kidney or liver gene expression. We investigate three chemicals in detail and provide plots for the rest in the supplement. In many cases, chemicals have well-known toxicity endpoints and biological mechanisms of response, which allows a comparison of our methodology to recognized biological activity. Our methodology is compared to that which is employed in the EPA Transcriptional Assessment Product (ETAP) which uses BMDExpress 2 to determine median gene set BMD and BMDL values as estimates of biological potency. The detailed procedure of BMDExpress 2 is provided in the supplement.

Algorithm 1 Compute BMD

Require: gene set $g = \{g_i\}_{i \in G}$

- 1: **for** iteration i in $1 : N_{iter}$ **do**
- 2: Gibbs sample β, Λ, τ
- 3: $\Sigma \leftarrow \Lambda \Lambda^\top + I\tau$
- 4: Center: $\widehat{X\beta} \leftarrow X\beta - X\beta(0)$
- 5: $M \in \mathbb{R}^{d \times d} \leftarrow (\widehat{X\beta}_g)^\top \Sigma_g^{-1} \widehat{X\beta}_g$
- 6: Set dose response $M(d) \leftarrow \text{diag}(M)$
- 7: Estimate test statistic T : Algorithm 2
- 8: **if** $M(d_{\max}) < T$ **then** No BMD found:
- 9: $BMD_i \leftarrow d_{\max}$
- 10: **else**
- 11: $BMD_i \leftarrow \arg \min_d s.t. M(d) > T$
- 12: **end if**
- 13: **end for**
- 14: **if** $\frac{1}{N_{iter}} \sum 1(BMD_i = d_{\max}) > 0.01$ **then**
- 15: No BMD proposed
- 16: **else**
- 17: Return $BMD^* \leftarrow$ 5th Percentile of BMD_i
- 18: **end if**

Algorithm 2 Compute Statistic with Null Correction

Require: Parameters $\tau, X, \beta, \Lambda, \eta$, gene set $g = \{g_i\}_{i \in G}$, BMR α

- 1: Compute all residuals $\epsilon = Y - X\beta - \Lambda\eta$
- 2: Fit splines to residuals, $\epsilon = X\beta_0 + \epsilon_0$
- 3: $\Sigma_0 \leftarrow I\tau$
- 4: Compute aggregate null for gene set centered to $X\beta_0(0)$, $M_0 = (\widehat{X\beta}_{g0})^\top \Sigma_{g0} (\widehat{X\beta}_{g0})$
- 5: Set null response $T^*(d) \leftarrow \text{diag}(M_0)$
- 6: Compute the $\chi^2_{n_g}$ quantile difference for the gene set size, $T_X = F_{\chi^2_{n_g}}^{-1}(0.98) - F_{\chi^2_{n_g}}^{-1}(0.78)$
- 7: Sample the dose-adjusted mean shift by correcting for the null variability, $T = T_X + T^*(d)$
- 8: Return T

Algorithm

Implementation

We fit each data set corresponding to each chemical using the model in Section 2. A total of 7000 MCMC samples were taken, disregarding the first 2000 as burn-in. We monitored all parameters, but focus on the BS-BMD estimates. We showcase the fenofibrate, furan, and methyleugenol results and leave the remaining chemicals to the supplement. While the parameter estimates vary across chemicals and tissues, the resulting interpretations carry over.

This section is organized as follows. We first present posterior BMD estimates across hallmark gene sets. Next, we illustrate how a single point from the posterior BMD distribution is sampled, with examples of a weak and strong signal corresponding to a high and low BMD. We then show a realization of the correlation structure when there are many signals in the data. We provide a table and scatter plot comparing our BMD estimates to those provided in Gwinn et al. [2020] and then provide chemical-specific commentary on the posterior BMD plots. Additional results, including BMD estimates for null data, comparison to BMD values derived from histopathology, and MCMC diagnostics for null data and results are provided in the supplement.

Posterior BS-BMD estimates

Figures 3, 4, and 5 show summary kernel density estimates of the posterior BS-BMD samples with a blue triangle at the 5th percentile and a green diamond at the mean. This value is similar to the BMD lower bound (BMDL) for a single endpoint, which is frequently employed in regulatory risk assessment. It encapsulates the variability of the experiment and estimation.

Diffuse densities suggest that the BS-BMD estimates varied widely across MCMC iterations, and the dose at which the response achieved the required BMR could vary greatly in each iteration of the MCMC routine, whereas narrow estimates suggest the BS-BMD is confidently specified. Figure 6 illustrates how the computation of the BS-BMD agglomerates for a single hallmark set in the case of a weak signal. The aggregated response of Equation (3) (in green) slightly exceeds the test statistic, shown as a dotted line, due to natural variability in estimation. While some genes in the set seem to have a large contribution, the aggregated dose response is not significant compared to the aggregated null response of Equation (4) (in orange) computed by the sampling procedure. The corrected aggregated response using Equation (5) (in blue) does not cross the threshold, so the maximum dose is provided as the BS-BMD. In other samples, the corrected response may still cross the threshold but at a higher dose. Figure 7 shows a single posterior estimates for a single hallmark set for a well defined BS-BMD. With such a strong signal, correcting for variability from the null response only shifts the BS-BMD across a small region. A few genes with large responses or many genes with moderate responses can both contribute to a gene set having a low BS-BMD with low estimated uncertainty. Note that the magnitude of the aggregated response depends on the number of genes and magnitude of individual responses in the set.

Covariance Estimates

Figure 8 shows a realization of the gene-by-gene correlation computed for the Fenofibrate liver responses of Figure 3 averaged across 100 MCMC samples. Each sample is converted to a covariance matrix first before averaging. The rows and columns (ie genes) are ordered using a hierarchical clustering with a distance given by 1 minus the correlation. The plot shows every fifth gene because details are too fine for the given resolution. Recall that the purpose of the correlation is to model how genes change in unison before considering exposure. Removing the latent factor allows us to work with decorrelated dose-responses and a well-defined test statistic.

In the figure, there are several layers of structure visible. There are approximately four large blocks and smaller, weakly correlated blocks suggesting additional processes activating or deactivating at higher doses. Note that the clusters do not cleanly map to Hallmark gene sets, so the term "processes" is used loosely. Off-diagonal regions can be negatively correlated, implying that the corresponding blocks of genes move in opposite directions in response to the increasing dose.

Comparison to BMDExpress PoDs

To investigate our approach, we compare the BS-BMD approach to PoDs calculated using BMDExpress. Table 1 provides a small selection relevant to the examples presented. A full comparison to all PoD results is given in the supplementary information. We provide the median and the lower 5th percentile of the posterior BS-BMD distribution and a rank ordering based on the 5th percentile. One can think of the

median as the central estimate of the PoD for the pathway and the lower bound as the PoD value that includes statistical uncertainties. Corresponding to these values are BMDEExpress computed values, where the central PoD estimate is computed from the median gene specific BMD for a given pathway, and the lower bound is taken from the median gene specific BMDL for the same pathway. This BMDEExpress pipeline mimics the EPA's current practice. The rank comparison illustrates that the BS-BMD method can correctly prioritize the hallmark sets, while the EPA method is more susceptible to false discoveries from noisy responses. Plots illustrating the entire ranking of hallmark sets are available in the supplement.

The BMDL comparison is visualized with a scatter plot in Figure 9. This plot summarizes and compares two data sets: the BMDEExpress results and the BMDs computed for apical endpoints, representing histopathological outcomes as the response rather than gene expression. For both comparisons we plot the lowest BMDL against the corresponding BS-BMDL estimate. This is an apples-to-oranges comparison because a change in gene expression may not be the cause of a pathological outcome, and the corresponding lowest BMDL may not even match the tissue of the pathology, which includes spleen and brain. For example, a compound like Ginseng was not found to cause any pathological outcomes, but responses in gene expression are detected in both the BS-BMD and EPA methods. Despite this issue, we find that the BS-BMDLs can be similar to the apical BMDs and occupies a middle ground between approaches.

Fenofibrate

Fenofibrate is a lipid-lowering medication primarily used to reduce cholesterol levels and triglycerides (<https://www.drugs.com/monograph/fenofibric-acid-fenofibrate.html>). Fenofibrate is a member of the fibrate class of drugs and works by activating peroxisome proliferator-activated receptor alpha (PPAR α). The nuclear receptor PPAR α , upon binding of fenofibrate, translocates to the nucleus and heterodimerizes with the retinoic acid receptor (RXR) to bind to its DNA-response element and regulates gene expression of multiple genes (including APOA 1/2/-V/AC3) that further results in changes to fatty acid metabolism and decreased circulating triglyceride levels via decreases in LDL expression, in LDL-C density, in LDL-C oxidation, and increases in lipoprotein lipase expression and lipoprotein clearance [Staels et al., 1998, House and Motsinger-Reif, 2020]. Figure 3 supports this notion with many strong kidney and liver signals. The potent responses are related to lipid and energy metabolism and appear in both kidney and liver: fatty acid metabolism, peroxisome, and adipogenesis, all of which are known effects of PPAR α activation [Varga et al., 2011, Goto et al., 2011]. The liver shows additional strong signals for fat related pathways, such as bile acid metabolism and cholesterol homeostasis, and oxidative stress or inflammation such as the interferon response pathways which are plausibly related to toxicological effects of fenofibrate at higher doses in rodents [Varga et al., 2011, Li et al., 2009]. Many of the other signals don't seem to have a clear connection to cholesterol and are plausibly secondary to the metabolic effects of fenofibrate [Corton et al., 2018].

Furan

Figure 4 shows samples from the posterior BS-BMD distribution for exposure to Furan. Furan is a volatile, colorless liquid often used in chemical manufacturing. If ingested or inhaled, furan is rapidly absorbed. In the liver it undergoes

metabolic activation to an epoxide form primarily via the enzyme CYP2E1 [Gates et al., 2012], which then reacts with and damages cellular macromolecules such as protein and DNA [Robles, 2014]. This leads to activation of cellular responses to counter the effects of the reactive chemical (e.g., up-regulation of glutathione pathways and unfolded protein responses). Once counter measures are exhausted cell death begins to occur [Carfagna et al., 1993] and the activation of damage response signaling in the form of dedifferentiation [Kotulkar et al., 2023], inflammation [Robinson et al., 2016] and regenerative processes [Gilgenkrantz and de l'Hortet, 2018] are activated. In agreement with the typically steep dose response seen with furan [Moser et al., 2009] a sizable number of the gene sets in liver are activate in narrow window (BMDL 2-8 mg/kg/day) with the gene sets related to cell cycle and regenerative/proliferation responses showing the highest potency (E2F Targets, G2M checkpoint, Mitotic Spindle, MYC Targets 1 and 2). This is likely due to onset of proliferative lesions after 5 days of exposure to furan. Proliferation associated gene set responses were closely followed in potency by gene sets related to cell death (apoptosis) and dedifferentiation (fatty acid and xenobiotic metabolism, both down-regulated upon review of the data). Other notable responses in the very narrow gene set response range are reactive oxygen species pathway (related to glutathione metabolism), unfolded protein response, UV response up/DNA repair/P53 response (a related to response to damage to DNA) and inflammatory response. Consistent with the known weaker toxicological effects of furan in the kidney [Von Tungeln et al., 2017], no gene sets were found to be active in this organ.

Methyleugenol

Methyleugenol, a natural compound found in various essential oils and commonly used spices such as nutmeg [Tan and Nishida, 2012], belongs to a subclass of phenylpropenes called alkenylbenzenes. A number of these compounds are known to be genotoxic and hepatocarcinogenic in rats [Eisenreich et al., 2021]. Methyleugenol undergoes bioactivation primarily in the liver and is metabolized by cytochrome P450 enzymes to form reactive metabolites, such as 1'-hydroxymethyleugenol [Gardner et al., 1997]. These metabolites are capable of causing oxidative stress, depletion of glutathione [Yao et al., 2016], formation of DNA adducts [Phillips et al., 1984], and elicitation of mutagenesis [Jin et al., 2013]. This leads to organ damage, which secondarily results in proliferative responses and inflammation. If these effects persist, they can lead to liver cancer. Based on what is known about the toxicological effects of methyleugenol in the liver, it is expected that the most potentially responding hallmark gene sets would be related to xenobiotic metabolism (response to the initial chemical insult), reactive oxygen species pathway (response to oxidative stress and glutathione depletion), P53 pathway (activated due to DNA damage), and E2F/Myc targets (cell proliferation). A review of the most potentially responding hallmark gene sets according to the BS-BMD analysis includes xenobiotic metabolism, P53 pathway, and Myc/E2F targets. Interestingly, there also appear to be potent effects on two related gene sets, fatty acid metabolism and peroxisome, which may indicate a stress-related shift in essential liver functions, potentially a byproduct of damage caused by methyleugenol [Tarlow et al., 2014].

Table 1. Comparison of our proposed PoD values to EPA equivalents for Fenofibrate, Furan, and Methyleugenol in the liver. Our method PoD refers to the median and BMDL refers to the 5th percentile of the posterior distribution, with similar quantities presented for the current EPA methodology. Note that rank is specified for the BMDL and may not match the POD rank.

Chemical	Hallmark	PoD	BS-BMDL	Rank	EPA PoD	PoDL	Rank	Active Genes
Fenofibrate	Fatty Acid Metabolism	0.92	0.68	1	12.43	7.86	6	57/72
Fenofibrate	Peroxisome	1.83	1.16	2	10.89	7.27	4	30/42
Fenofibrate	Xenobiotic Metabolism	2.36	1.59	6	23.66	14.34	7	49/94
Fenofibrate	Adipogenesis	2.42	1.55	4	24.67	15.20	9	52/63
Fenofibrate	Oxidative Phosphorylation	2.43	1.48	3	22.72	14.49	8	37/40
Furan	E2f Targets	2.40	1.90	1	3.74	2.86	5	65/94
Furan	G2m Checkpoint	2.59	2.08	3	3.75	2.84	4	52/84
Furan	Mitotic Spindle	2.69	1.98	2	3.50	2.78	3	25/43
Furan	Myc Targets V1	3.12	2.40	5	4.13	3.33	10	42/55
Furan	Mtorc1 Signaling	3.41	2.39	4	6.18	4.68	21	51/98
Methyleugenol	Xenobiotic Metabolism	116.57	67.16	1	121.23	94.15	10	28/94
Methyleugenol	Fatty Acid Metabolism	138.50	88.80	6	160.20	123.28	17	29/72
Methyleugenol	P53 Pathway	141.49	91.80	7	128.80	93.95	9	25/85
Methyleugenol	Peroxisome	149.64	67.60	2	116.78	88.91	6	14/42
Methyleugenol	Mtorc1 Signaling	156.10	86.94	5	167.26	125.91	19	30/98

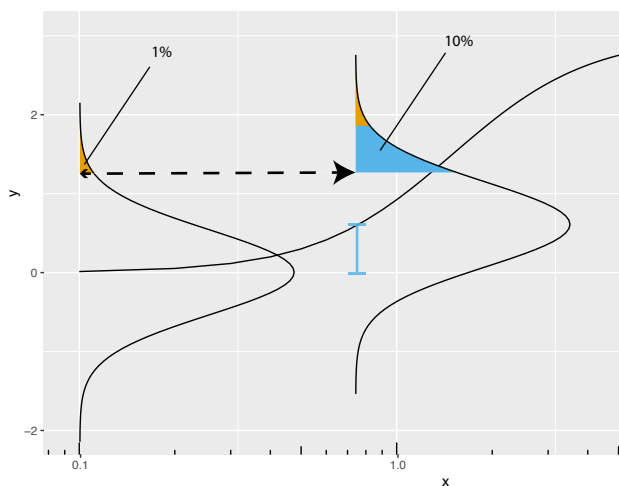


Fig. 1. The probabilistic method for determining BMD based on Crump [1995]. Under the assumption of Gaussian error, the top 1% in orange is fixed as the cutoff for an adverse response, where the cutoff is illustrated with the dotted arrow. The BMR is set to 10% and the BMD is the dose where the change in response yields the BMR change above the cutoff. The change in response is equal to the difference of quantiles.

Null Data Sets

We applied our procedure to null data that maintained the gene-to-gene correlation. Essentially all hallmark sets report BS-BMDs equal to the maximum dose, so plots are left to the supplement.

Figures

Discussion

The BS-BMD method provides a principled, statistical approach to determining if a gene set has experienced a sufficiently large aggregate change in response, where “large” is determined by the choice of the BMR. Our technique is a direct, multivariate extension of the now standard approach pioneered by Crump [1995] that underpins the common practice of using one standard deviation as the BMD, which is used

as the PoD. Our results suggest that using known, curated pathways such as the Hallmark Gene Ontology can produce values that are consistent with prior knowledge of chemical effects and reasonably close to apical data or existing methods in use by the EPA.

However, computing a BMD does not provide the same information as an apical endpoint study. Genes may increase or decrease in expression without an observable cellular or pathological response. This implies that our method cannot be applied without a biologist or toxicologist to determine if the findings are relevant or to classify the expression change as dangerous or harmless.

Furthermore, the aggregation process of our technique sacrifices the direction of response to avoid having effects that cancel out, so it is not possible to say if the gene set as a whole has been up-regulated or down-regulated. Predicting this direction in addition to the BMD is an area of future work that would require highly detailed gene expression network models such as the causal analysis of Krämer et al. [2014]. Post-hoc analyses are also possible. While our BS-BMD is sampled within the MCMC iteration to obtain posterior distributions, the BMD or other quantities such as direction of regulation could be computed using the posterior parameter and covariance estimates instead.

The results with null data suggest that our method has a false discovery rate of 0. The practical effect of this can be seen in the rankings shown in table 1, where the most relevant hallmark sets were not given the highest rank by the EPA method. A false discovery rate of 0 may inflate the missed discovery rate, but testing for the true false and missed discovery rates is non-trivial because the simulated data needs to represent gene expressions with realistic correlations in addition to realistic responses. Compounding the issue is that the gene sets provided by prior pathway knowledge, such as the hallmark gene sets, do not necessarily correspond to fixed, well quantified mechanisms that show consistent and uniform responses across all tissues. For example, as shown in our comparison to EPA methods, we find only a fraction of the genes in a set are active. Such results could suggest corrective mechanisms or nonlinear responses within the pathway that vary with tissue, but the verification of pathways and simulation thereof is beyond the scope of this work.

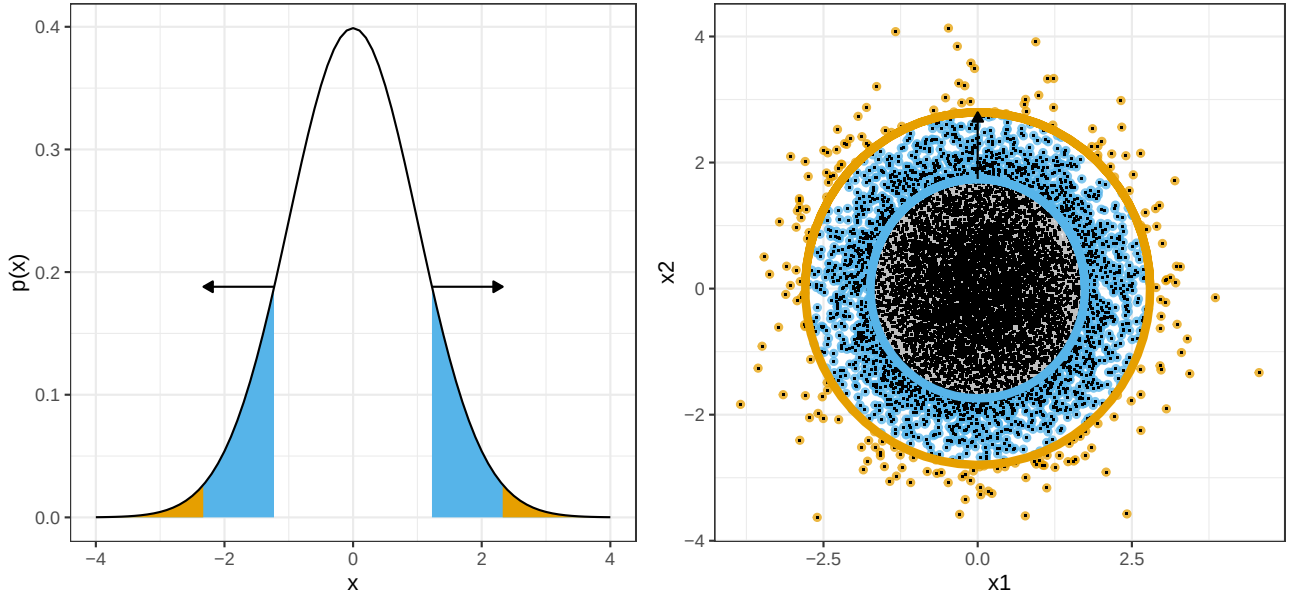


Fig. 2. The multivariate extension introduces a norm that requires accounting for both upper and lower tails. The cutoff for the test statistic in the multivariate case can be visualized as the difference between two shells.

Even without the full mechanistic knowledge or direction of expression or clear link to pathology, our method provides great value in terms of hypothesis generation. Given a high throughput study, a scientist can apply our technique to sort the pathways from most to least affected by a particular chemical and have more confidence that the highest ranked pathways are meaningful signals compared to existing methods. This can help determine follow-up work such as gene knockout studies that can lead to a better mechanistic understanding of the chemical effect. Such studies could in turn better inform the gene sets of relevance, creating a positive feedback loop around our method that can iteratively reveal mechanisms of toxicity.

Appendix

Bayesian Model Implementation Details

The full hierarchical model is

$$Y_i = X\beta_i + \Lambda_i\eta + \epsilon_i$$

$$\beta_i | c_i \sim N(0, I c_i^{-2}), \quad \epsilon_i | \tau_i \sim N(0, I \tau_i^{-2})$$

$$\eta_{ij} \sim N(0, 1) \quad \Lambda_{jh} | \gamma_h, \phi_{jh} \sim N(0, \phi_{jh}^{-1} \gamma_h^{-1})$$

Hyperpriors are specified with priors:

$$c_i \sim \Gamma(1, 1), \quad \tau_i \sim \Gamma(2, 2) \quad \phi_{jh} \sim \Gamma(v/2, v/2)$$

$$\gamma_h = \prod_{\ell=1}^h \delta_\ell, \quad \delta_1 \sim \Gamma(a_1, 1), \quad \delta_\ell \sim \Gamma(a_2, 1)$$

The covariate matrix X is generated from a B-spline basis with four basis functions evaluated at the observed doses. For creating figures with smooth curves, the basis functions are re-evaluated at a finer scale over the dose range. Since the basis functions are not changed, the fitted coefficients can be used to plot the fixed effects or estimate where the test statistic crosses the BMR-based threshold.

MCMC Diagnostics

We give evidence that our MCMC fitting procedure rapidly converges to the posterior. Given the model's complexity, we emphasize that the model parameters, especially the latent factor terms, can appear to not converge but still result in stationary distributions for the computed test statistics. This is often due to rotational symmetries and individual model parameters being non-identifiable; consequently, we focus our diagnostics on identifiable quantities that have actual meaning (e.g., correlation, and the BS-BMD).

Figures 10 and 11 show trace plots of the benchmark doses for two hallmark sets. The BMD reaches stationarity within 1000 iterations or the BMD values concentrate on the maximum dose with occasional samples appearing in the dose range. The supplement shows the Gelman-Rubin diagnostic for stationarity for all the parameters and suggests that the posterior samples are stationary. The results are based on 7000 posterior samples. Assuming a burn-in of 2000 as presented, the effective sample size for the BMD estimates for each gene set vary from 24% to 65%, meaning the 5000 MCMC samples corresponds to between roughly 1200 and 3200 independent samples. When there is no signal, the posterior sample consists of just the maximum dose and the effective sample size is not well defined.

Probabilistic Formulation

Here we review the derivation of Crump [1995] from a probabilistic perspective and observe that BMR is defined as both a “fold-change” scalar and as a change in the probability of an adverse response in that work. Let the cumulative distribution function for the response R given a centered response be denoted by F_R and the inverse CDF by F_R^{-1} . By definition, we have

$$P(R > F_R^{-1}(0.99)) = 0.01$$

The method of Crump [1995] is then to find a threshold H where the adverse response has increased by a specified Benchmark

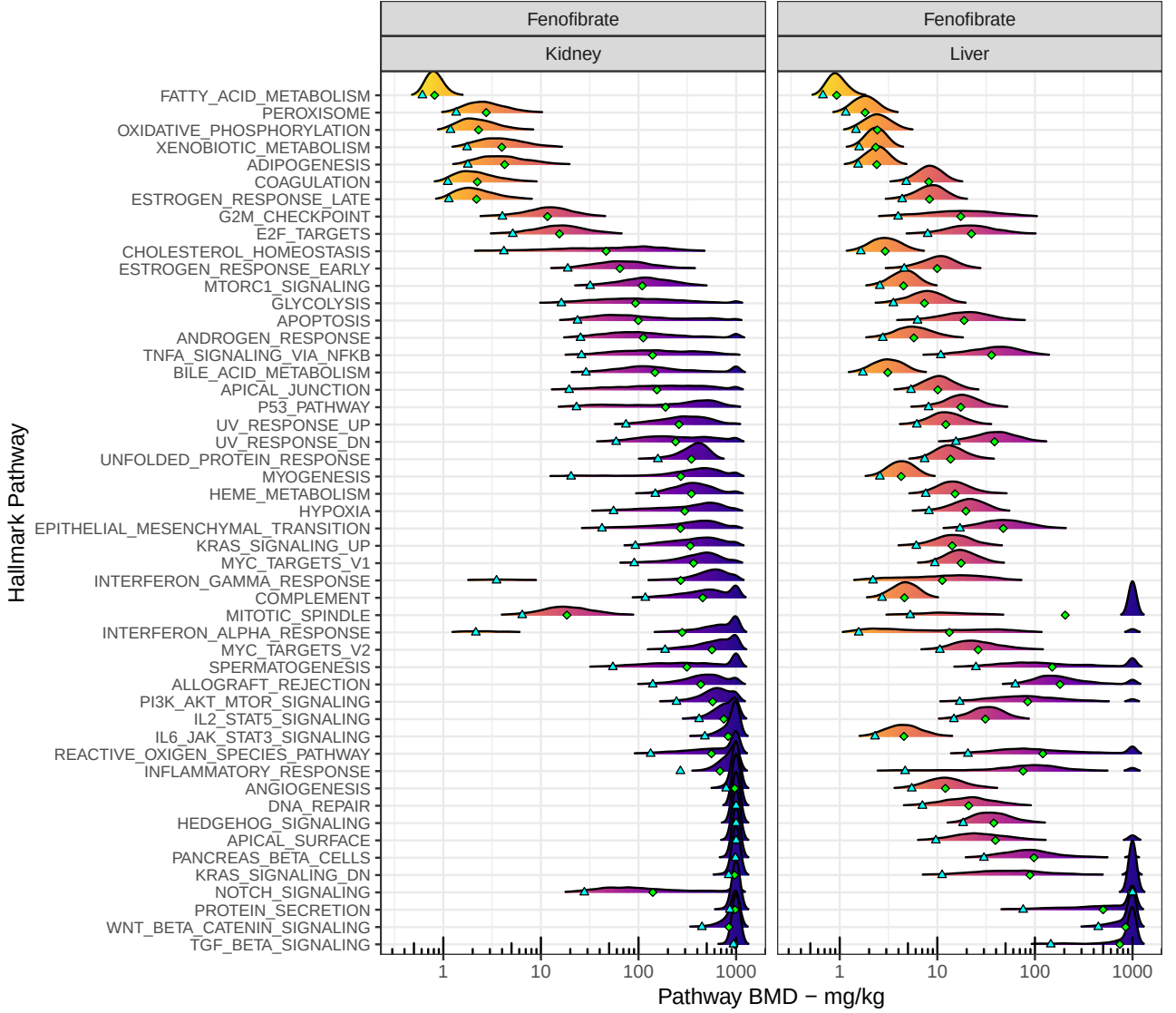


Fig. 3. Fenofibrate BS-BMD simulations by each Hallmark pathway for kidney (left) and liver (right). The distribution of 5,000 posterior samples for the BS-BMD (X-Axis) are shown for each Hallmark gene set (Y-axis) as well as the median (diamonds), and 5th percentile (triangles). The density is shown when the probability mass is at least 0.01.

Response (BMR) such as 10%,

$$P(R + H > F_R^{-1}(0.99)) = 0.01 + \text{BMR} = 0.11$$

Moving the constant H to the right side, we have

$$P(R > F_R^{-1}(0.99) - H) = 0.11$$

An immediate solution is to let $H = F_R^{-1}(0.99) - F_R^{-1}(0.99 - \text{BMR})$, which is approximately the 1.1 value mentioned earlier when $F = \Phi$, the Gaussian CDF. This expression is easily generalized to higher dimensional cases by substituting for the new random variable of interest. For example, aggregating with squared, normalized terms such as $\hat{R} = \sum_i R_i^2 / \sigma_i^2$ yields the same basic calculation,

$$P(\hat{R} + H > F_R^{-1}(0.99)) = 0.01 + \text{BMR}. \quad (6)$$

As before, we can use the trivial result $H = F_R^{-1}(0.99) - F_R^{-1}(0.99 - \text{BMR})$. Using a 10% change in adverse response for the BMR will not recover identical statistics for all transformations, so care is needed to match the effect of the univariate case. Additionally, if γ is the extreme tail probability of interest, then this generalizes to $H = F_R^{-1}(1 - 2\gamma) - F_R^{-1}(1 - 2\gamma - 2\text{BMR})$, for our approach. As described in the main text, the quadratic response formula requires doubling the probability mass to account for the negative tail being folded into the positive tail. This procedure assumes that the error distribution does not change with the center, which is not generally true. However, the observed doses may not be sufficiently dense to provide error distributions near the true BMD, so the assumption of equal variance (homoskedasticity) is a reasonable simplification and alternative to attempting to interpolate the error distribution between doses.

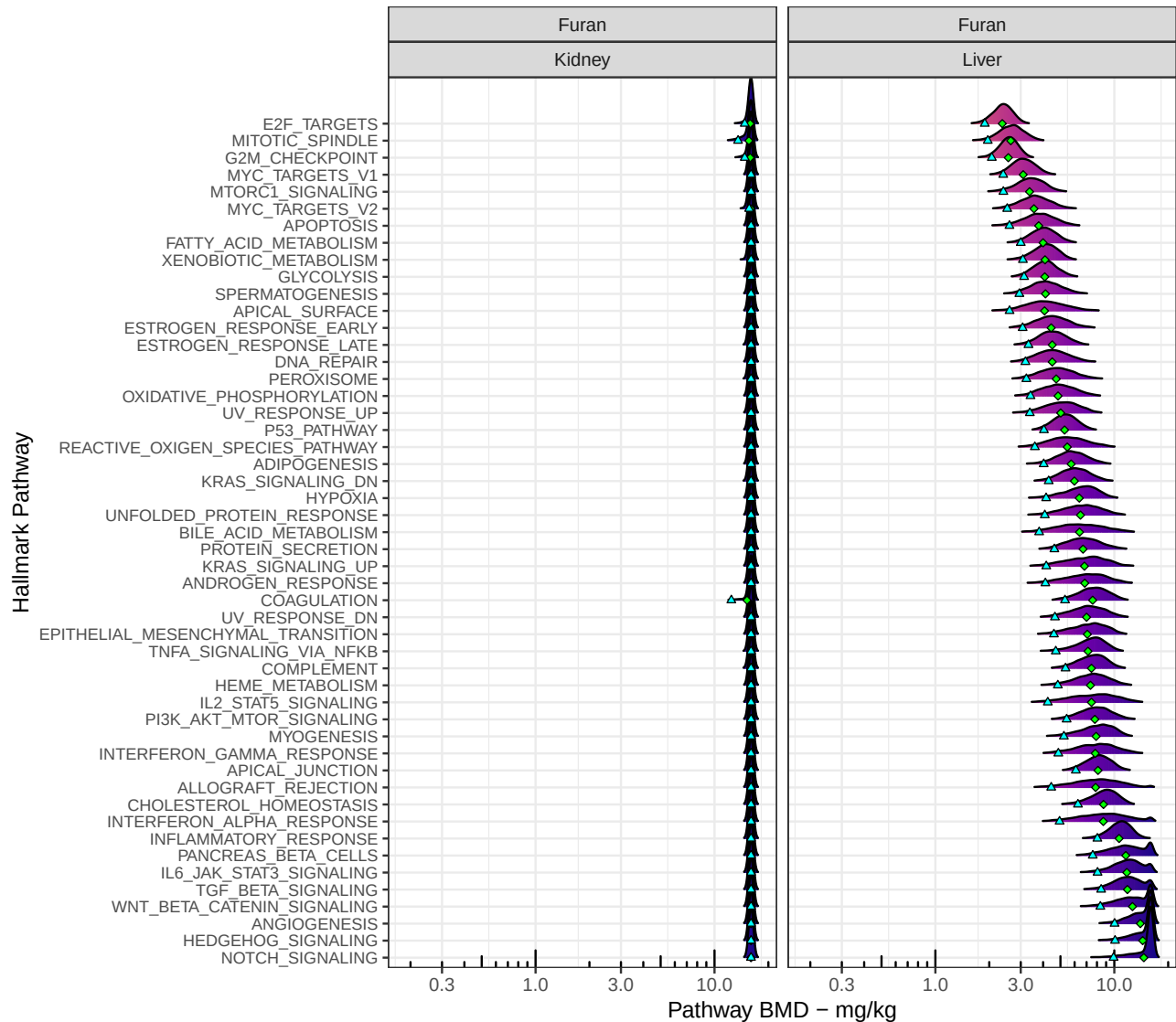


Fig. 4. Furan BS-BMD simulations by each Hallmark pathway for kidney (left) and liver (right). The distribution of 5,000 posterior samples for the BS-BMD (X-Axis) are shown for each Hallmark gene set (Y-axis) as well as the median (diamonds), and 5th percentile (triangles). The maximum dose is returned as the BMD when the response is not significant, so the kidney shows no active pathways. The density is shown when the probability mass is at least 0.01.

Competing interests

No competing interest is declared.

Author contributions statement

D.Z. and M.W. conceived the mathematical basis for the methodology, D.Z. and M.W. implemented the methodology. S.A., J.H., F.P. and K.M. analyzed the results. D.Z., M.W., S.A., J.H., F.P. and K.M. wrote and reviewed the manuscript.

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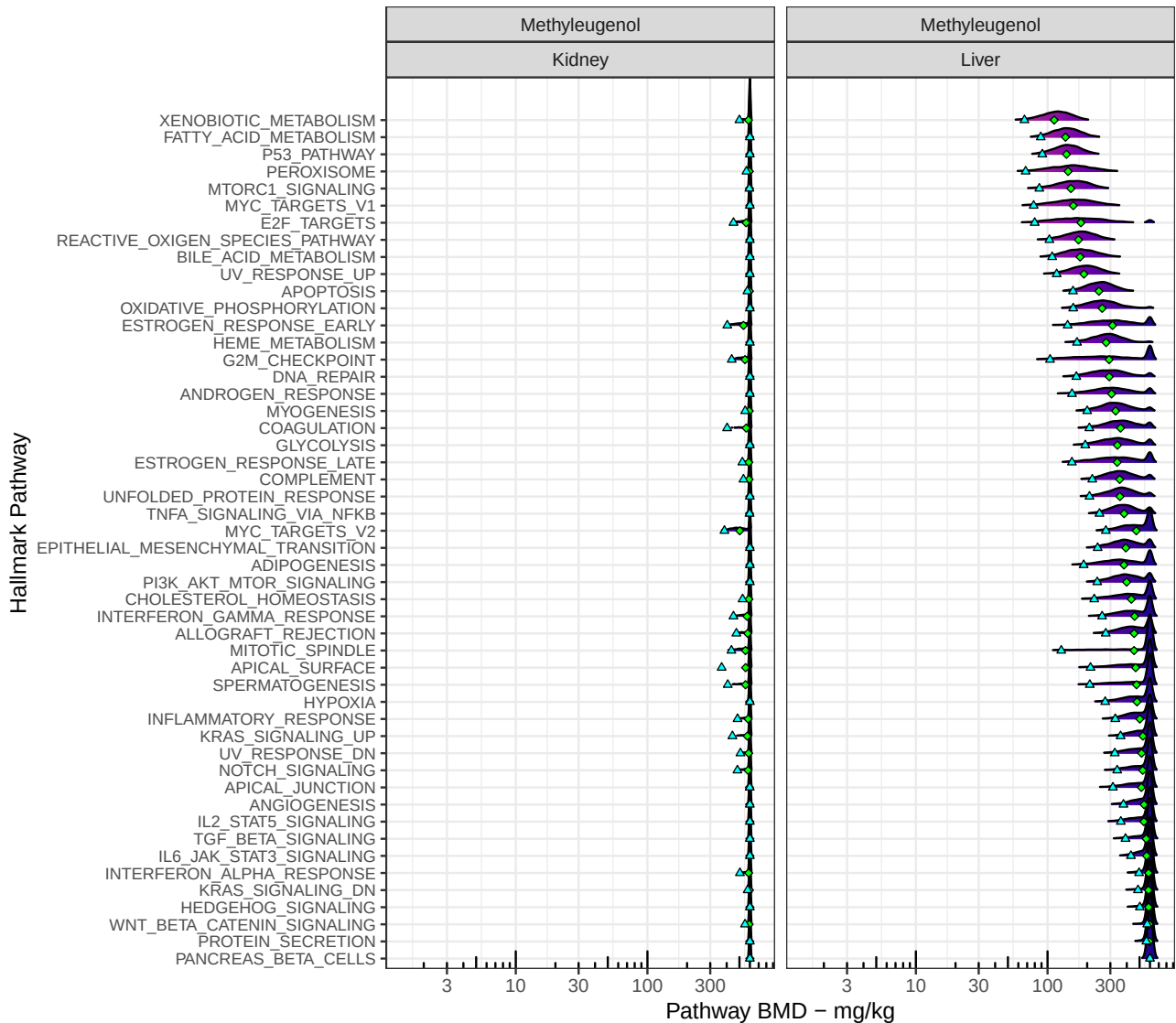


Fig. 5. Methyleugenol BS-BMD simulations by each Hallmark pathway for kidney (left) and liver (right). The distribution of 5,000 posterior samples for the BS-BMD (X-Axis) are shown for each Hallmark gene set (Y-axis) as well as the median (diamonds), and 5th percentile (triangles). The maximum dose is returned as the BMD when the response is not significant, so the kidney shows no active pathways. The density is shown when the probability mass is at least 0.01.

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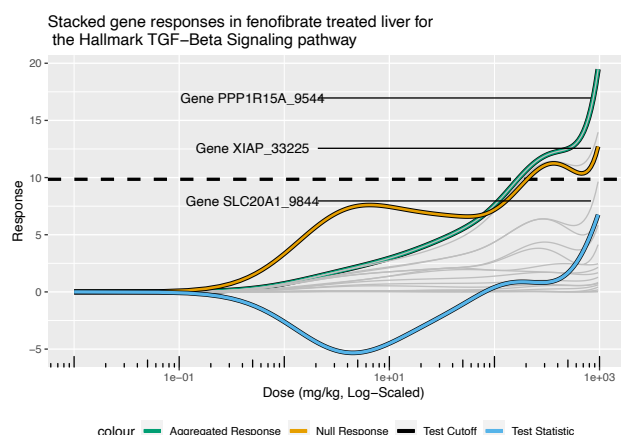


Fig. 6. Weak signal in the liver with Fenofibrate. There are 17 genes in the TGF Beta Signaling set and the difference between the 98th and 78th quantile in the noise distribution is roughly 10, represented by the black dotted line. The aggregate effect in green of all the genes has a moderate BMD before correcting for the spline model variance in orange. After correcting, the test statistic in blue does not cross the cutoff, so the BMD is set to the maximum dose. In some cases the test statistic still crosses the cutoff, suggesting a true effect but at a higher dose. The top three genes in the stack are shown with the top gene coinciding with the aggregate effect but the top genes vary with each iteration.

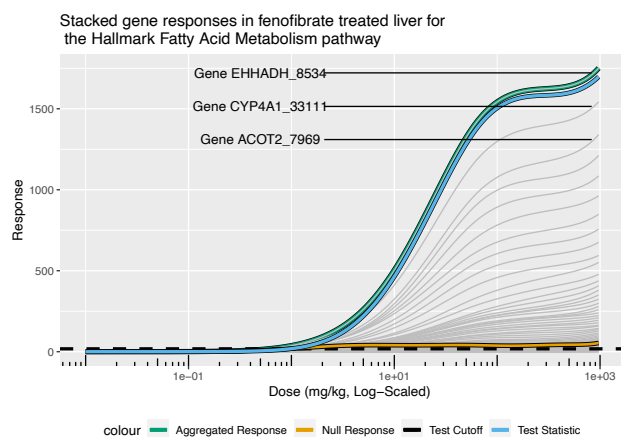


Fig. 7. Strong signal in the liver with Fenofibrate. The Fatty Acid Metabolism hallmark set has 72 genes, resulting in a population noise distribution with larger mean and spread. The difference in response between the 98th and 78th quantiles is roughly 125, represented by the black dotted line. The aggregated dose response in solid black easily exceeds the limit and there are many genes that contribute to the overall response. The top three genes are shown and do not significantly vary across iterations.

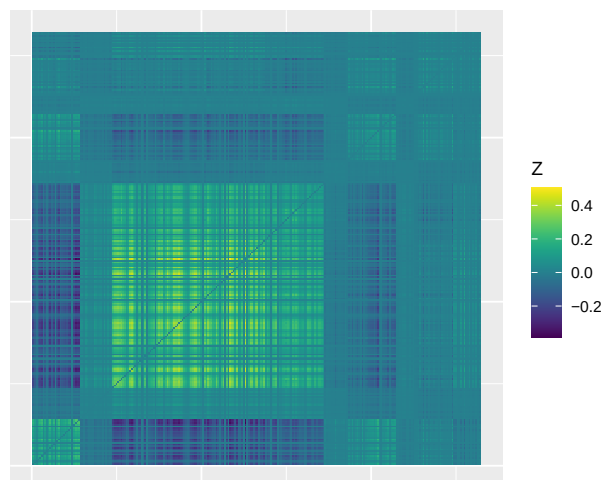


Fig. 8. Correlation matrix of genes in the liver exposed to fenofibrate. Genes are correlated through latent factors rather than directly through the response, so this does not represent an empirical covariance matrix. The covariance is computed by averaging over 100 posterior samples. The rows and columns (i.e., genes) are ordered using a hierarchical clustering with a distance given by 1 minus the correlation, but only every 5th gene is shown because the details are too fine to print.

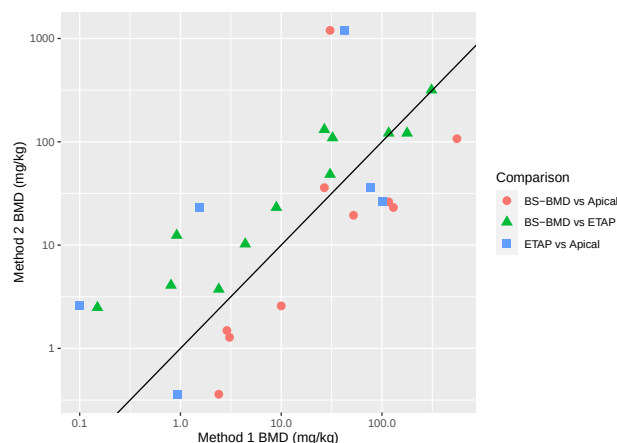


Fig. 9. Three-way comparison of BMD estimates from the EPA methodology and apical study to our BS-BMD method using a scatter plot and paired results. BS-BMD tends to return larger values than the apical study and smaller values than the EPA method. The points correspond to the lowest BMD values reported for the EPA or apical study, if present, across all tissues.

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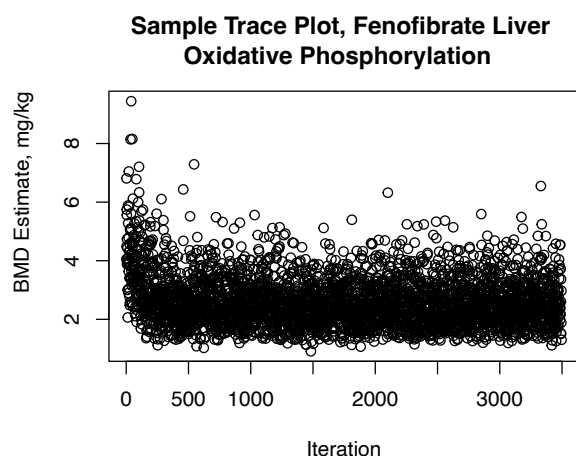


Fig. 10. Trace plot showing sampled values for the BMD for the oxidative phosphorylation set under Fenofibrate in the liver. Only half of the 7000 sampled points are shown for clarity. Stationarity is achieved within 1000 iterations

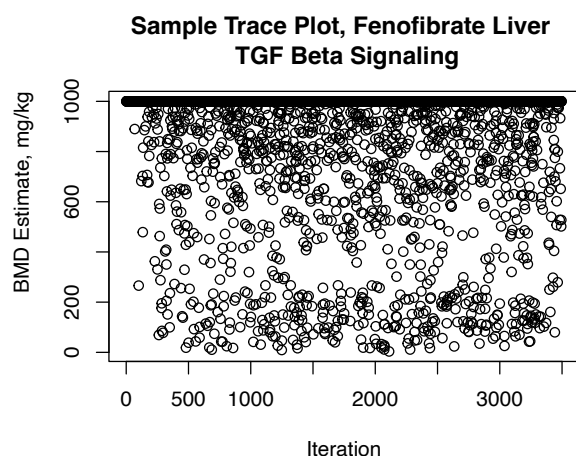


Fig. 11. Trace plot showing sampled values for the BMD for the TGF Beta signaling gene set under Fenofibrate in the liver. Only half of the 7000 sampled points are shown for clarity. The concentrated mass on the maximum dose is typical for gene sets with weaker signals.

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